Werkstudent Coding Challenge: Wazuh

1. Objective

Build a small SOC workflow focusing on three modules:

- Triage
- Incidents

This challenge evaluates your ability to design, implement, and document a practical system.

2. Timeline

You have 1 week to complete the assignment.

We expect a functional, not a production-ready system.

3. Recommended Stack

Backend: Node.js (TypeScript preferred)

Frontend: React

Database: MongoDB or PostgreSQL or SQLite

API: REST

You may use mock JSON data instead of integrating real systems.

4. Deliverables

- Complete source code repository
- Implementation Guide document
- Setup instructions
- Sample/seed data
- Optional: Docker, tests, screenshots

5. Module Requirements

You must implement the following modules:

5.1 Triage Module (Mandatory)

Purpose: Analyst work queue for reviewing alerts.

Display:

- Alert list (ID, timestamp, severity, rule, host, IP, status)
- Filters and sorting
- Alert detail view with raw JSON
- Actions: mark true/false positive, ignore, escalate

Implementation:

- Backend endpoints for alerts and triage decisions
- Store triage state separately
- Frontend alert list + detail UI

5.2 Incidents Module (Mandatory)

Purpose: Group alerts into cases.

Display:

- Incident list (ID, title, severity, status, alert count)
- Incident detail with timeline, alerts, comments

Implementation:

- Endpoints for incidents and linking alerts
- Database tables for incidents and comments
- UI for creating and managing incidents

5.3 Detection Module (Optional but preferrable)

Purpose: Analyze rule performance and noise.

Display:

- Rule ID, description, severity
- Alert count, false positive %, escalation %
- Noisy rule identification

Implementation:

- Aggregate alerts by rule
- Combine with triage decisions
- Provide simple noise logic

6. Implementation Guide (Mandatory)

Your submission must include a guide covering:

- Project overview
- Architecture
- Setup instructions
- Module details
- Assumptions
- Limitations
- Future improvements

7. Submission Checklist

- Code repository included
- Application runs locally
- All min. first 2 or max all 3 modules implemented
- Implementation guide included
- Sample data included
- README with setup steps